Вариан 15 [redacted]

$F = 3x_1 + x_2 + 3x_3 + 2x_4 \to \max$

$2x_1 + x_2 + 2x_3 + x_4 \leq 20$

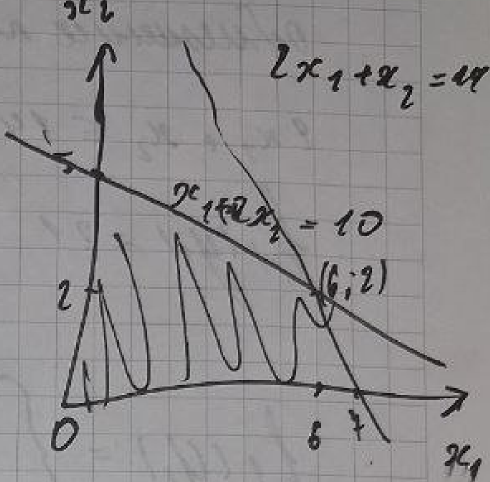
$2x_1 + x_2 \leq 14$

$x_1 + 2x_2 \leq 10$

$3x_3 + x_4 \leq 18$

$x_3 + 2x_4 \leq 11$

$x_j \geq 0, j = \overline{1,4}$



Блок 1

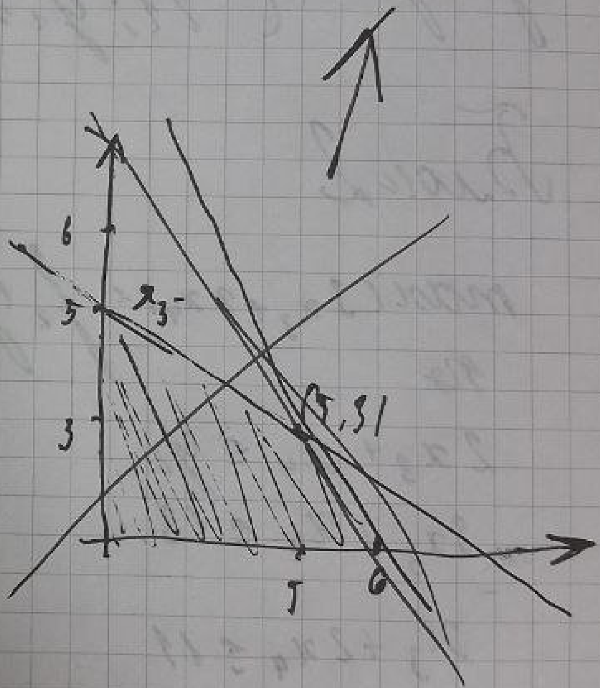
$\max(3x_1 + x_2) = f_1(y_1)$

$2x_1 + x_2 \leq y_1$

$2x_1 + x_2 \leq 14$

$x_1 + 2x_2 \leq 10$

$x_1 \geq 0, x_2 \geq 0, y_1 \geq 0$



а) $0 \leq y_1 \leq 14$:

$x_1^0 = \frac{y_1}{2}$; $x_2^0 = 0$

$f_1(y_1) = \frac{3y_1}{2}$

б) $y_1 \geq 14$

обмеження перестає діяти, оскільки:

$2x_1 + x_2 \leq 14$

$f_1(y_1) = 21$

$$f_1(y_1) = \begin{cases} \frac{3}{2}y_1; & 0 \leq y_1 \leq 14 \\ 21; & y_1 \geq 14 \end{cases}$$

Блок 2

$\max(3x_3 + 2x_4) = f_2(y_2)$

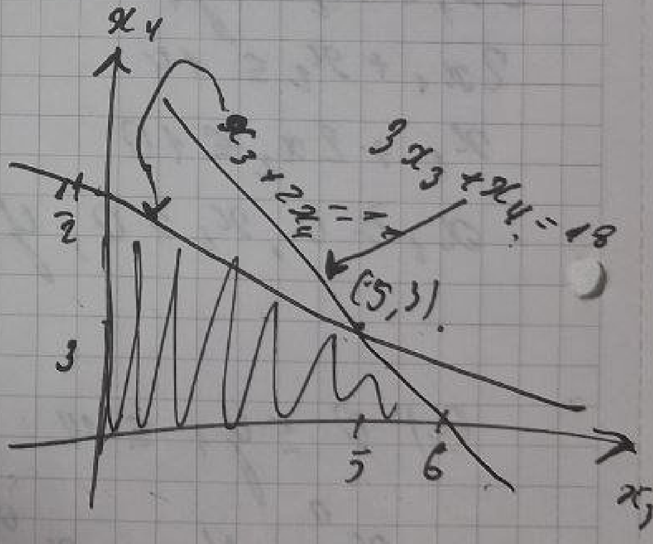
за

$2x_3 + x_4 \leq y_2$

$3x_3 + x_4 \leq 18$

$x_3 + 2x_4 \leq 11$

$x_3 \geq 0, x_4 \geq 0, y_2 \geq 0$



~~а) $0 \leq y_2 \leq \frac{11}{2}$:~~

~~$x_3^0 = 0$~~

~~$x_4^0 = y_2$~~

~~$f_2(y_2) = 4y_2 + 11$~~

~~Т. (5,3): $y_2 = 13; f_2(y_2) = 21$~~

а) $0 \le y_2 \le \frac{11}{2}$:

$x_3^0 = 0$

$x_4^0 = y_2$

$f_2(y_2) = 2y_2$

б) $y_2 \geq \frac{11}{2}$

$2x_3 + x_4 = y_2$

$x_3 + 2x_4 = 11$

$x_3 = \frac{2y_2 - 11}{3}$

$x_4 = \frac{22 - y_2}{3}$

$f_2(y_2) = \frac{4y_2 + 11}{3}$

Т. (5,3): $y_2 = 13; f_2(y_2) = 21$

$$f_2(y_2) = \begin{cases} 2y_2; & 0 \le y_2 \le \frac{11}{2} \\ \frac{4y_2 + 11}{3}, & 11 \le y_2 \le 13 \\ 21; & y_2 \geq 13 \end{cases}$$

Координуюча задача

$\max F(y) = f_1(y_1) + f_2(y_2)$

$0 \le y_1 + y_2 \le 20$

1. $\frac{df_1}{dy_1} = \frac{3}{2}$; $0 \le y_1 \le 14$

$\frac{df_2}{dy_2} = 2$; $0 \le y_1 \le \frac{11}{2}$

$\frac{df_2}{dy_2} > \frac{df_1}{dy_1} \Rightarrow y_2 = \frac{11}{2}$

2. $\frac{df_1}{dy_1} = \frac{3}{2}$; $0 \le y_1 \le 14$

$\frac{df_2}{dy_2} = \frac{4}{3}$; $\frac{11}{2} \le y_2 \le 13$

$\frac{df_1}{dy_1} > \frac{df_2}{dy_2} \Rightarrow y_1 = 14$

3. $\frac{\partial f_1}{\partial y_1} = 0 ; \; y_1 \geq 14$

$\frac{\partial f_2}{\partial y_2} = \frac{4}{3} ; \; \frac{11}{2} \leq y_2 \leq 13$

$\frac{\partial f_2}{\partial y_2} > \frac{\partial f_1}{\partial y_1} \Rightarrow y_2 = 20 - 14 = 6$

$x_1^0 = \frac{14}{2} = 7$

$x_2^0 = 0$

$x_3^0 = \frac{2 \cdot 6 - 11}{3} = \frac{1}{3}$

$x_4^0 = \frac{22 - 6}{3} = \frac{16}{3}$

Перевірка

$2 \cdot 7 + 0 \cdot 2 + \frac{1}{3} + \frac{16}{3} = 14 + 6 = \underline{20}$

$F_{max} = 3 \cdot 7 + 0 + 3 \cdot \frac{1}{3} + 2 \cdot \frac{16}{3} = 21 + 1 + \frac{32}{3} =$

$= \underline{\frac{98}{3}}$